

Integration of Traffic Signal Control and Connected and Automated Vehicle Speed Planning for Energy-Efficient Mobility

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37 Abstract

38 Urban traffic congestion is a primary driver of global energy inefficiency, yet
39 mitigation via connected and automated vehicles has remained largely theoretic-
40 al. We bridge this gap with a real-world integrated framework that coordinates
41 predictive traffic signal control with longitudinal CAV planning. The framework
42 combines traffic signal control with a roadmanship-aware eco-driving speed plan-
43 ning algorithm, while a low-level model predictive control module ensures safe
44 and smooth execution under real vehicle dynamics. A rigorous multi-platform
45 evaluation through microscopic simulation, vehicle-in-the-loop testing, and on-
46 road experiments demonstrates reduced stop frequency, smoother trajectories,
47 and energy savings of up to 20.16%, providing practical insight into deploying
48 integrated traffic signal control and CAV speed planning strategies to enhance
49 both vehicle energy efficiency and traffic flow in mixed traffic environments.

50 **Keywords:** Eco-Driving, Signal Control Optimization, Real-World Deployment, Mixed
51 Connected and Automated Traffic Environment, Energy-Efficient Mobility Systems

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Introduction

Transportation plays a vital role in the American economy, which accounted for 6.7% of GDP in 2022 [1]. However, the current transportation system is quite energy-intensive. Based on U.S. Energy Information Administration published data, about 27% of total U.S. energy consumption in 2021 was used for transportation [2]. The increased demand for transportation also causes traffic congestion, especially on urban signalized arterials. To improve transportation efficiency, the U.S. Department of Energy (DOE) and the U.S. Department of Transportation have supported various research and development efforts to create new tools and technology solutions that increase mobility efficiency. Among those technologies, connected and automated vehicles (CAVs) and adaptive traffic signal control are two of the most promising solutions. CAVs can obtain real-time information of surrounding vehicles, road information, and traffic signal phase and timing (SPaT) via vehicle-to-vehicle and vehicle-to-infrastructure (V2I) communications. With this information, CAV technologies have great potential to reduce congestion, save energy, and improve safety through different vehicle control strategies. Meanwhile, various advanced adaptive traffic signal control algorithms have been proposed and deployed to optimize traffic flow by adjusting traffic signal timings based on real-time traffic conditions collected by various sensors.

Traffic signal optimization and vehicle speed optimization were usually studied separately by researchers in transportation engineering and automotive engineering. In the transportation field, various traffic signal control methods have been studied to improve the control performance of signalized intersections [3], including actuated [4] and adaptive (planning-based) traffic signal control [5, 6]. A traditional fixed-time traffic signal control operates with predefined phase sequences and durations based on historical traffic data, regardless of real-time conditions. In contrast, an adaptive traffic signal control dynamically adjusts signal timings and phases in response to current traffic states, typically leading to more efficient intersection management. An adaptive traffic signal control predicts future traffic states within a prediction horizon and generates optimal signal timing plans to meet control objectives and constraints. To find optimal traffic signals, various solution approaches have been applied, such as dynamic programming (DP) [7], agent-based solutions [8, 9], neural networks [10, 11], reinforcement learning [12, 13], and graph reinforcement learning for large-scale networks [14–16]. DP offers precise control policies for well-defined systems, whereas agent-based approaches provide flexibility in modeling decentralized and interactive behaviors. Neural networks excel at capturing complex, nonlinear relationships in traffic dynamics. Reinforcement learning enables adaptive decision-making based on real-time feedback, and graph reinforcement learning further enhances scalability and spatial coordination in large, networked traffic systems. In addition, emerging technologies such as vehicular communications [17] and CAVs [18] have been increasingly leveraged to improve traffic signal control performance. By enabling real-time V2I communication, CAVs can transmit detailed traffic data—such as vehicle speed, position, and heading—to traffic signal controllers. This rich, real-time information allows for

more responsive and adaptive traffic signal control strategies, leading to enhanced traffic flow efficiency and reduced delays compared with those of traditional sensor-based approaches.

Frequent stop-and-go driving, traffic congestion, and poor road conditions can significantly increase vehicle fuel consumption [19]. CAV technologies are being investigated to reduce fuel consumption, including predictive vehicle dynamics and powertrain control [20, 21], eco-routing [22], eco-approach and departure at signalized intersections [23, 24], platooning [25, 26], and cooperative adaptive cruise control [27, 28]. In frequent stop-and-go driving, a significant amount of fuel is consumed because of unnecessary braking and subsequent accelerations. Optimal velocity trajectory control aims to avoid stopping at intersections or to decelerate gradually by coasting before arriving at a site where braking would otherwise be required, providing significant fuel savings and improving traffic flow. Energy-efficient vehicle velocity optimization for connected vehicles at signalized intersections has been investigated by many researchers [29]. Dong et al. proposed an eco-driving control strategy for vehicles approaching a signalized intersection [30]. An energy-efficient driving strategy was generated by an iterative DP algorithm to improve vehicle energy efficiency and overall traffic throughput. The speed-planning algorithm considers vehicle queues at the intersection. The optimal speed profile is followed by a proportion-integration controller with consideration of collision avoidance constraints. Li et al. proposed a deep reinforcement learning-based reference speed generation for CAVs to minimize fuel consumption while satisfying the constraints of traffic lights and safety rules [31]. The multiobjective optimization function of the eco-driving control problem is transformed into the value function of the deep reinforcement learning algorithm by designing fuel, traffic light, and safety rewards. Dong et al. proposed a hierarchical control strategy for CAVs to generate optimal velocity profiles [32]. The upper-level controller generates an efficient speed profile with the consideration of queue effects and multiple traffic lights. The fuel-optimal reference speed profile is solved by the direct multiple shooting algorithm and is tracked by a lower-level controller with predictive cruise control. Dong et al. proposed a two-level eco-approach and departure strategy for CAVs passing through multiple intersections [33]. The upper-level algorithm calculates a series of green windows for each intersection using an A* algorithm. The objective is to reduce travel delays. The lower-level algorithm optimizes the speed trajectory using an iterative DP algorithm to minimize energy consumption and battery life losses. More recently, Hesami et al. proposed a bi-layer eco-driving control architecture for autonomous electric vehicles at signalized intersections, in which an upper layer plans energy-efficient reference trajectories and a lower layer enforces them via real-time longitudinal control [34]. While their work demonstrates substantial energy savings, the evaluation is limited to simulation-based scenarios without explicit consideration of mixed traffic or communication uncertainties.

CAV technology advancement enables real-time information exchange between traffic signal controllers and CAVs, **providing opportunities** to further improve traffic and vehicle control performance at intersections. Research on integrated optimization of traffic signal timings and vehicle speed profiles at intersections is gaining increasing attention. Du et al. proposes a coupled vehicle-signal control method to

optimize traffic signal timing and driving trajectories of CAVs to improve traffic efficiency and vehicle fuel economy [35]. Signal timing is optimized to minimize the total delay at the intersection. CAV eco-driving trajectories are generated using the signal timing information and planned arrival time to reduce fuel consumption. Only simulation validation is performed without field implementation of traffic signals and vehicle controllers. Tajalli and Hajbabaie present a cooperative signal timing and vehicle trajectory optimization method at intersections [36]. The joint signal timing and trajectory control is formulated as a mixed-integer nonlinear program in a single optimization problem. Although the presented method shows a reduction in average travel time and fuel consumption compared with that of an existing signal timing and trajectory control approach, this centralized approach impedes the distributed vehicle control that is crucial for scalability when the number of CAVs increases in the traffic flow. Long et al. propose an integrated optimization of traffic signals and vehicle trajectories [37]. The proposed method is evaluated by a microscopic simulation for an intersection with four approaches. Guo et al. use DP to optimize a signal timing plan that minimizes intersection delays and employs a shooting heuristic (SH) algorithm for planning CAV trajectories to reduce fuel consumption [38]. Moreover, the proposed DP-SH algorithm is a centralized approach, and numerical experiments are conducted to examine the algorithm’s performance. Hou et al. propose a cooperative and integrated vehicle and intersection control for energy efficiency [39]. A DP algorithm is designed to find the optimal signal timing and vehicle schedules with minimum energy loss. A model predictive control (MPC) strategy is adopted to ensure that vehicles pass through the intersection according to the schedules generated by DP. An isolated two-way intersection is modeled using VISSIM traffic simulation software. The vehicle speed profiles generated by VISSIM are simulated in the Future Automotive Systems Technology Simulator (FASTSim) [40] to calculate vehicle fuel economy. Recent studies have further advanced eco-driving strategies for automated or connected vehicles in the presence of signalized intersections and mixed traffic environments. For example, Ding et al. (2024) proposed an intersection eco-driving strategy that jointly optimizes traffic signal control and vehicle trajectory planning under mixed traffic conditions [41]. Their method demonstrates that cooperative coordination between signals and vehicles can reduce delays and improve energy efficiency compared with non-cooperative approaches. However, the approach assumes idealized communication between vehicles and infrastructure, and does not incorporate real-world uncertainties such as sensor noise, V2X latency, or heterogeneous driver behaviors. At the system level, integrating CAVs into urban traffic environments offers notable socioeconomic benefits by reducing congestion and improving fuel efficiency. Through vehicle-to-vehicle and V2I communication, CAVs help smooth traffic flow and minimize delays, resulting in lower fuel consumption [42, 43].

Although recent studies on integrated traffic signal timing and vehicle speed control have made important progress, most existing frameworks remain structurally different from the deployable architecture developed in this study. Prior centralized joint optimization methods [35–39] generally optimize traffic signal timing and CAV trajectories in simulation, often for isolated intersections, and do not explicitly evaluate subsystem communication, vehicle actuation dynamics, or field implementation constraints. More

recent studies, including Zou et al. [44], Ding et al. [41], Liu et al. [45], and Li et al. [46], further advance eco-driving and mixed-traffic signal-vehicle coordination, but their evaluations remain simulation-based or near-real-world and do not demonstrate full closed-loop signal-vehicle coordination under live traffic with real vehicle hardware. In contrast, the proposed framework combines three deployable components: a receding-horizon predictive bilinear signal controller that extends the single-cycle formulation of Wang et al. [47] to optimize signal timings over multiple future cycles; an onboard roadmanship-aware eco-driving (RAED) planner [48] that incorporates predicted queue length and a distance-dependent minimum-speed constraint to maintain socially compatible behavior in mixed traffic; and a cloud-based real-time mobility control and communication system that enables closed-loop signal-vehicle coordination under practical communication conditions. These structural differences lead to measurable performance benefits across multiple evaluation stages. Therefore, the contribution of this study is not limited to real-world deployment, but lies in the integrated design and validation of a multi-cycle predictive infrastructure controller, a roadmanship-aware vehicle planner, and a cyber-physical evaluation framework spanning simulation, vehicle-in-the-loop (VIL) laboratory testing, and on-road experiments.

The major contributions of this study include the following:

1. A deployable integrated signal-vehicle coordination framework is developed by coupling multi-cycle predictive bilinear traffic signal control with onboard RAED speed planning for CAVs in mixed traffic.
2. A cloud-based real-time mobility control and communication system (RTMCS) is implemented to support real-time exchange of signal timing, queue, and vehicle-state information between CAVs and transportation infrastructure under practical communication conditions.
3. A comprehensive multiplatform evaluation is conducted through calibrated microscopic simulation, VIL laboratory testing with real vehicle dynamics, and on-road field demonstration, quantifying both signal-level benefits and vehicle-level energy-saving performance.

The remainder of this paper is organized as follows: Section 2 presents the experimental results obtained through simulation, VIL testing, and on-road demonstration; Section 3 describes the methodological framework, including bilinear traffic signal control, vehicle control algorithms, and the cyber-physical infrastructure supporting real-time deployment; and Section 4 concludes the study and discusses the implications of the findings as well as future research directions.

Results

System Overview: A Closed-Loop Framework for Cyber-Physical Integrated Control

We present a comprehensive, real-time integrated control framework that harmonizes transportation infrastructure with CAV dynamics to achieve energy-efficient mobility in mixed traffic environment. As illustrated in Figure 1, the system architecture functions as a unified cyber-physical system driven by two primary agents: a predictive traffic signal controller and a CAV planner. Unlike traditional systems where vehicles

238 react passively to static traffic signal cues, our framework establishes a continuous
 239 data exchange that enables optimization of both signal timing and vehicle trajectories.

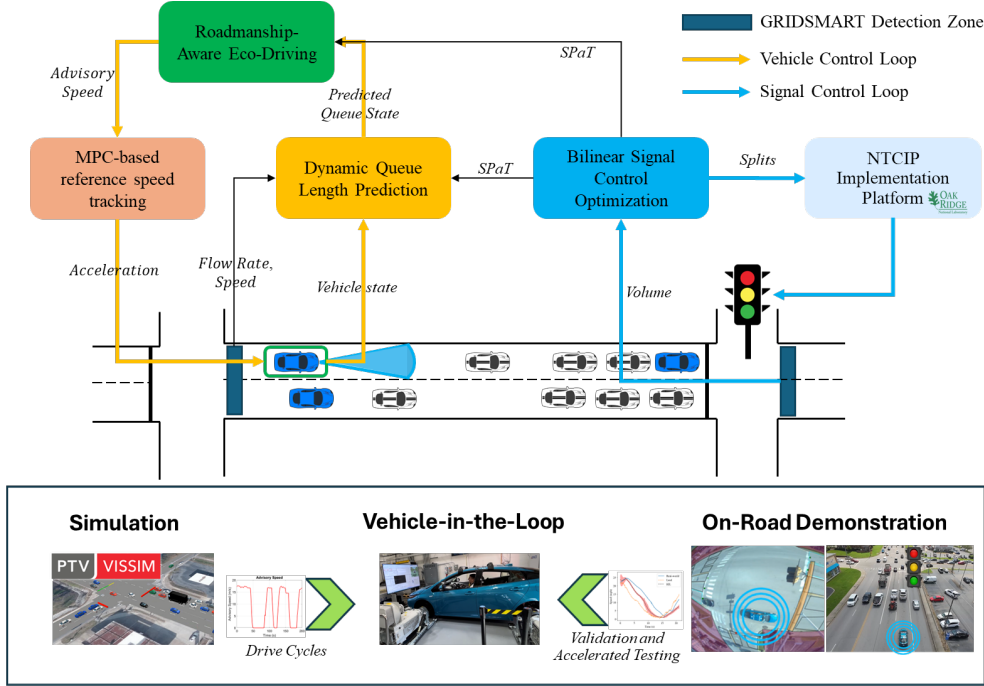


Fig. 1: An overview of the integrated traffic signal and CAV speed planning framework.

240 At the infrastructure level, we deploy a Predictive Bilinear Traffic Control Algo-
 241 rithm. This module processed real-time high-resolution traffic flow data captured by
 242 vision-based sensors, e.g., GridSMART to dynamically optimize traffic signal phase
 243 splits. Compared with conventional actuated systems which rely on past traffic condi-
 244 tions, the proposed controller uses a bilinear state-space model to predict traffic states
 245 over a rolling horizon of multiple future cycles. The predictive capability allows the
 246 system to maximize **intersection throughput** while simultaneously generating reliable
 247 SPaT forecasts. These forecasts are broadcast via the cloud-based RTMCS, which
 248 leverages the Intelligent Transportation System Protocol (NTCIP) standard to ensure
 249 seamless communication.

250 On the vehicle side, the architecture excutes a hierachical control strategy des-
 251 ignated as Roadmanship-Aware Eco-Driving (RAED) [48]. The READ algorithm
 252 actively ingests the broadcasted SPaT data and estimated queue length to compute
 253 an energy-optimal velocity trajectory. We introduce the term "Roadmanship-Aware"
 254 to define a novel constraint layer: the planner optimizes for energy efficiency subject
 255 to social compatibility constraints, ensuring that the CAV maintains headways and
 256 velocities that align with the flow of surrounding human-driven traffic. This high-level

trajectory is then executed by a low-level Model Predictive Control (MPC) module. The MPC layer translates the advisory speeds into precise longitudinal throttle and braking commands.

The integration of these subsystems creates a synergistic feedback loop. The signal controller adjusts timings to maximize traffic throughput and clear downstream queues, while the CAV adjusts its arrival time to coincide with predicted green windows, thereby minimizing energy-intensive stop-and-go events. Queue prediction is integrated directly into the speed planning process, serving as a critical environmental variable that allows the vehicle to anticipate blockages. To rigorously validate this architecture, we implemented a multi-stage evaluation pipeline with microscopic simulation (VISSIM), VIL laboratory testing, and a live deployment on public roads in Chattanooga, Tennessee.

Multiplatform Evaluation Framework

Evaluation is a crucial step in validating the proposed algorithm’s performance across various environments, ensuring its robustness and effectiveness in real-world applications. In this study, the team developed a comprehensive evaluation framework to assess the integrated traffic-vehicle control system through three distinct stages: (1) microscopic traffic simulation, (2) VIL testing in the connected and automated vehicle environment (CAVE) laboratory at Oak Ridge National Laboratory (ORNL), and (3) on-road experiments. Figure 2 summarizes the strengths and weaknesses of different stages and their evaluation applicability.

Stage	Strength	Weakness	Evaluation Capability		
			Traffic Signal	Vehicle Control	Integrated Control
Simulation	➤ Controlled & repeatable environment	➤ Ignores real-world complexity	✓	✓	✓
	➤ Large sample size (millions of scenarios)	➤ No real vehicle dynamics (physics model only)			
VIL	➤ Real vehicle dynamics (Chassis/Engine)	➤ Resource-intensive (Requires lab + car)	●	✓	●
	➤ Repeatable environment	➤ Sensor spoofing complexity			
On-Road	➤ Real traffic & communication flows	➤ Uncontrolled & unrepeatable	✓	●	●
	➤ Absolute physics fidelity	➤ Safety risks for edge cases			

Fig. 2: Summary of different experimental stages and their evaluation applicability.

Stage 1: Statistical Validation via Microscopic Simulation. Stage 1 involves using detailed simulation models to replicate traffic flow and vehicle behavior on a granular level. The advantage of microscopic traffic simulation is that it allows for controlled, repeatable experiments in which numerous variables can be adjusted to evaluate the proposed algorithm’s performance in diverse scenarios. However, a limitation is that

it may not fully capture the complexities of real-world driving conditions, such as unpredictable human behavior or environmental factors.

Stage 2: Powertrain Dynamics Verification (VIL). Stage 2 involves integrating a real vehicle with a virtual environment in the CAVE laboratory. This setup enables a more realistic testing environment by providing real-time feedback and interaction with the algorithm. The advantage of VIL testing is that it bridges the gap between simulation and real-world conditions, offering a more immersive and accurate assessment of the proposed algorithm’s performance. The drawback, however, is that it can be resource-intensive and may not fully replicate every aspect of real-world driving, particularly in terms of unpredictable external elements.

Stage 3: Field Deployment Feasibility via On-Road Experiment. Stage 3 involves deploying the algorithm in actual vehicles on public roads to observe its performance in real-world conditions. The primary advantage of on-road field experiments is that they provide the most accurate validation of the proposed algorithm’s effectiveness and safety in everyday driving scenarios, including dealing with unpredictable elements such as human drivers and varying environmental conditions. The main drawbacks are the high cost, potential safety concerns, and logistical challenges associated with real-world testing, as well as difficulty controlling experimental variables.

The Simulation stage is effective for evaluating all of the control systems. It provides a controlled environment with calibrated real-world scenarios for extensive testing, making it highly suitable for evaluating complex control algorithms before moving to more resource-intensive stages. The VIL stage, well-suited for evaluating vehicle control but limited for signal and integrated control systems, integrates real vehicle dynamics into a controlled environment, offering a middle ground between simulations and real-world tests. Although VIL addresses some simulation limitations such as vehicle dynamics, it remains resource-intensive and still does not fully capture real-world traffic complexities. The On-Road stage, suitable for traffic signal control but less effective for vehicle control and integrated control systems, provides the most realistic evaluation environment, incorporating real vehicle and traffic flow dynamics as well as real communication and localization. Despite this stage’s high resource demands and potential safety concerns, it is crucial for validating the practical implementation of control systems. However, the uncontrolled nature of on-road tests makes them less suitable for evaluating vehicle and integrated control systems under consistent conditions.

Simulation-Based Evaluation

This section presents the evaluation results of integrated control of traffic signals and CAVs obtained from the simulation environment. The simulation environment represents a 1.6-mile segment of the Shallowford Road corridor in Chattanooga, Tennessee, containing eight signalized intersections. Traffic demand was constructed using 15-minute turning-movement counts collected by GRIDSMART vision sensors during the morning peak hour (8:00–9:00 a.m.). The demand data were used to define vehicle flow inputs and turning ratios for the baseline scenario. The baseline signal operations were configured using the morning peak timing plans supplied by the City of

Chattanooga. All eight intersections operate with a 100 s cycle length and coordinated phasing along the corridor. Figure 3 shows the corridor location and a scaled schematic of the eight signalized intersections, including intersection numbering, the eastbound (EB) and westbound (WB) through-movement directions, and the lateral spacing between adjacent intersections (panels a-b), and summarizes the AM-peak signal timing parameters used in the baseline simulation, including signal offsets and major/minor green-time ranges for each intersection (panel c). These morning peak timing parameters were directly implemented in VISSIM to replicate real-world semi-actuated signal operations during the time period. Environmental conditions such as grade, weather, and pavement state were held constant in the simulation environment to isolate the performance of the integrated control strategies.

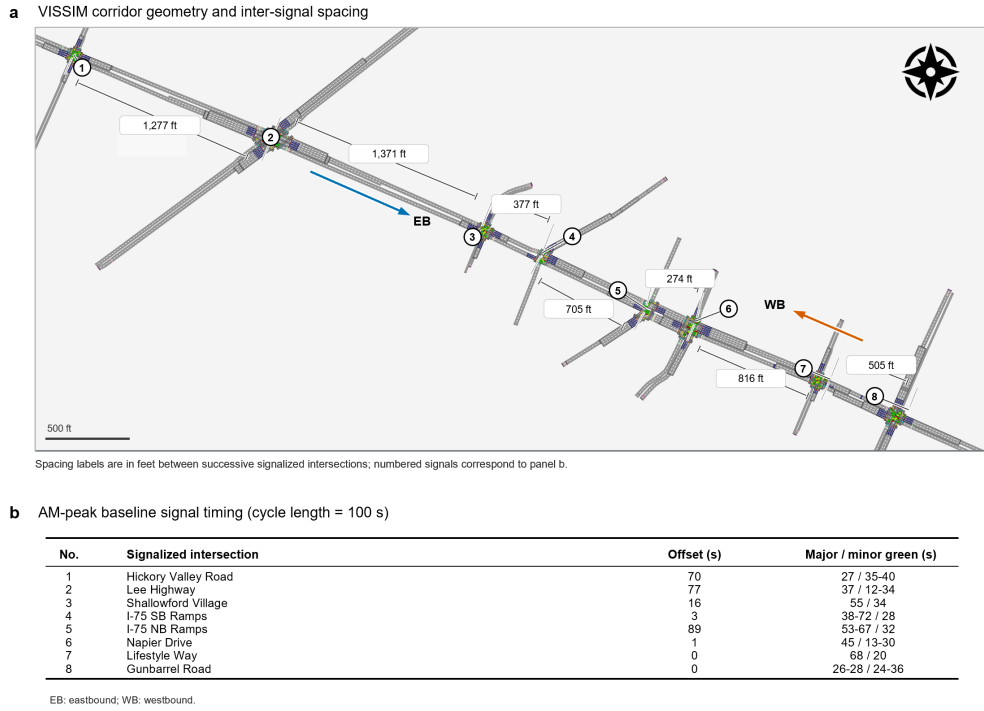


Fig. 3: Shallowford Road corridor network. (a) Map context showing the corridor location in Chattanooga, TN. (b) Scaled corridor schematic showing the eight signalized intersections (numbered 1–8), the eastbound (EB) and westbound (WB) through-movement directions, and the lateral spacing between adjacent intersections. (c) AM-peak signal timing parameters used in the baseline VISSIM simulation, including signal offset, major/minor green-time ranges, and distance from Intersection 1 (Hickory Valley Road), for a common cycle length of 100 s.

The simulation was calibrated using real-world traffic volume and speed data to ensure that the modeled traffic conditions were representative of observed corridor operations. Calibration was conducted for the morning peak, using field data collected on September 15, 2021. Traffic volumes were calibrated using the Geoffrey E. Havers (GEH) statistic [49], with at least 85% of turning movements required to satisfy $GEH < 5$. Space-mean speeds on selected roadway segments were calibrated using HERE speed data, with a mean absolute percentage error below 30% for the majority of segments. Maximum queue lengths were assessed through qualitative comparison to field observations. Across all time periods, the calibrated VISSIM models met standard calibration criteria recommended by FHWA [50]. Overall, the calibrated simulations were deemed suitable for evaluating the performance of the proposed traffic signal control and CAV speed planning framework.

In the simulation environment, the advisory speed trajectory for the ego vehicle was generated by our RAED speed planner and passed to VISSIM through the COM external controller interface. VISSIM’s simplified vehicle dynamics model was then responsible for translating the advisory speed into acceleration and deceleration behavior within the traffic stream. It is worth noting that, this internal model only provides a behavioral representation of vehicle motion and does not simulate high-fidelity physical or powertrain dynamics. To ensure consistency with the VIL and on-road experiments, the energy consumption cost function used by the RAED speed planner incorporated the mass and road-load coefficients (f_0, f_1, f_2) of the 2017 Toyota Prius Prime.

To ensure statistical robustness, all performance results were aggregated over 10 independent simulation runs for each scenario using different random seeds to account for stochastic variability in traffic flow and vehicle interactions. In total, 16 scenarios were evaluated: one baseline scenario without external vehicle or traffic signal control, in which vehicles followed the VISSIM internal driver model and traffic signals operated using replicated real-world semi-actuated controllers, and 15 integrated traffic-vehicle control scenarios spanning five market penetration rates (MPRs) and three levels of queue-length prediction ($5 \times 3 = 15$). These 15 scenarios are hereafter referred to as the integrated scenarios. In all integrated scenarios, traffic signals were controlled using the predictive bilinear signal control algorithm. The five different MPRs are 20%, 40%, 60%, 80%, and 100%. The three types of queue effect considerations are (1) no queue prediction, (2) implementable queue prediction based on emulated real-time GridSMART data, and (3) ideal queue prediction with assumed full knowledge of every vehicle’s real-time location and speed. The ideal queue prediction case is included solely as an upper-bound benchmark to contextualize the performance of the implementable queue estimation method, rather than as a realistic deployment assumption. Each scenario lasts 5,400 simulation seconds, with the first and last 900 s excluded from the evaluation.

To improve statistical transparency, simulation results are reported using both distributional and mean-precision summaries. For each integrated scenario, route-matched CAV energy-efficiency improvements were aggregated across 10 independent random-seed runs. Figure 4 is organized as a four-panel layout: the top panels (a, b) display box plots of energy-efficiency improvement distributions for the EB and WB

directions, respectively, where each box represents the interquartile range, the center line represents the median, whiskers represent the non-outlier range, and individual points denote outliers; the bottom panels (c, d) display the mean energy-efficiency improvement as a function of CAV market penetration rate for EB and WB, respectively, with error bars representing the 95% confidence interval of the mean. Together, these panels quantify both the stochastic variability in energy-efficiency improvement and the statistical precision of the reported mean estimates across the 10 random-seed runs.

Because the interaction between traffic signal control and traffic dynamics will be quite different across different scenarios, conducting an “apples-to-apples” comparison by simply picking the same vehicle identification (ID) is difficult. In this context, a pairwise route-based energy evaluation, based on the identification of vehicle pairs between the baseline scenario and each integrated traffic-vehicle control scenario that traveled the same route, was proposed in this study.

Vehicle pairs between the baseline scenario and each integrated scenario (e.g., baseline and 20% CAV, baseline and 40% CAV) were based on predetermined matching criteria (i.e., the traveled route should be the same, the time entering the simulation network should be within 10 s, and the difference between the travel distance should be within 0.2 mi). It is worth noting that only the CAVs from each integrated scenario were matched because the objective is to evaluate how the CAVs performed compared with the non-CAVs in this section. Figure 4 presents the energy savings for the CAVs in various integrated scenarios, comparing scenarios without queue prediction and with queue prediction for the eastbound (EB; panels a and c) and westbound (WB; panels b and d) directions. The results show that scenarios with queue prediction consistently achieve higher energy-efficiency improvement than those without, across all tested CAV market penetration rates. In addition, the energy-efficiency improvement for WB traffic is slightly higher than for EB traffic, likely because the WB corridor is more congested and the integrated control algorithm can avoid more stops under such conditions.

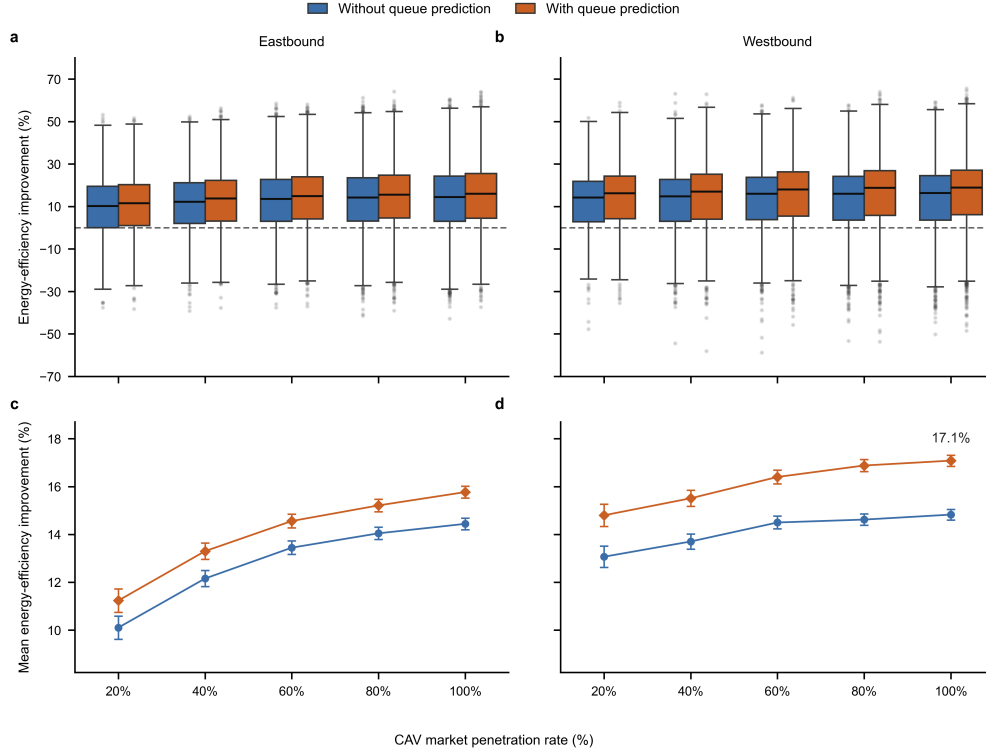


Fig. 4: Route-based comparison of route-matched energy-efficiency improvement for EB and WB directions. Top panels (a, b): box plot distributions for EB and WB, respectively, where boxes show the interquartile range, center lines show the median, whiskers show the non-outlier range, and points show outliers. Bottom panels (c, d): mean energy-efficiency improvement with 95% confidence intervals as a function of CAV market penetration rate (MPR) for EB and WB, respectively.

Among all tested scenarios, the 100% CAV scenario with queue prediction achieves the highest mean energy-efficiency improvement: 15.8% in the EB direction (Figure 4c) and 17.1% in the WB direction (Figure 4d). In both directions, incorporating queue prediction into the integrated traffic-vehicle control algorithm consistently and significantly increases the energy-efficiency improvement, confirming that integrating a queue length prediction algorithm is valuable for real-world deployment.

VIL Assessment

VIL tests were performed to evaluate the performance of integrated traffic-vehicle control system in the CAVE laboratory at ORNL. This laboratory is a virtually and physically integrated environment in which the energy benefit of CAVs can be evaluated using real hardware in virtual traffic conditions. The CAVE laboratory enables

comprehensive evaluations of CAV technologies through the cosimulation of real vehicles, traffic, virtual environments, hardware controllers, and communications [51]. In this study, the purposes of the VIL tests are to evaluate the energy benefits of the integrated traffic-vehicle control system with real vehicle dynamics and to verify if the vehicle is able to track the desired speed profiles generated in the VISSIM simulation. During VIL tests, the desired speed profiles are provided to the test vehicle. The low-level speed tracking controller uses MPC to generate optimal acceleration/deceleration commands to track the desired speed profile and minimize the acceleration/deceleration and jerk of the vehicle. The optimal acceleration/deceleration commands are then sent to the vehicle through dSPACE MicroAutoBox II to control the vehicle on the dynamometer in the lab.

Test scenarios were selected based on the following factors: vehicle entering time, CAV penetration rate, semi-actuated/predictive bilinear traffic signal control, with/without RAED speed planning, and with/without queue prediction. The drive cycles used in the VIL experiments were extracted directly from the calibrated simulation results described in Section 2.3. Each drive cycle represents a realistic vehicle trajectory along the Shallowford Road corridor under a specific control configuration. For each test group, three matched drive cycles were selected from the baseline scenario, the integrated scenario without queue prediction, and the integrated scenario with queue prediction, respectively. This matched design ensures that differences observed in vehicle behavior and energy consumption are attributable to the integrated control system rather than variations in traffic demand or environmental characteristics.

Because of limited resources, 4 groups of scenarios (i.e., 12 drive cycles, each group having 3 matched drive cycles) with different penetration rates have been tested on the VIL platform. During VIL tests, real-time vehicle signals from the vehicle controller area network (CAN) were recorded for the evaluation of battery consumption and speed tracking performance.

Figure 5 shows the space-time diagram of the three matched scenarios with the ego vehicles traveling on the same route and entering the network at the same time. The CAV penetration rates of both integrated scenarios are 80%. The surrounding vehicles are also plotted to illustrate overall traffic conditions. Figure 5 clearly shows that the integrated traffic-vehicle control can smoothen the movement of the vehicle through the corridor and reduce the number of stops for the ego vehicle. Moreover, integrating the queue length prediction can further improve the smoothness.

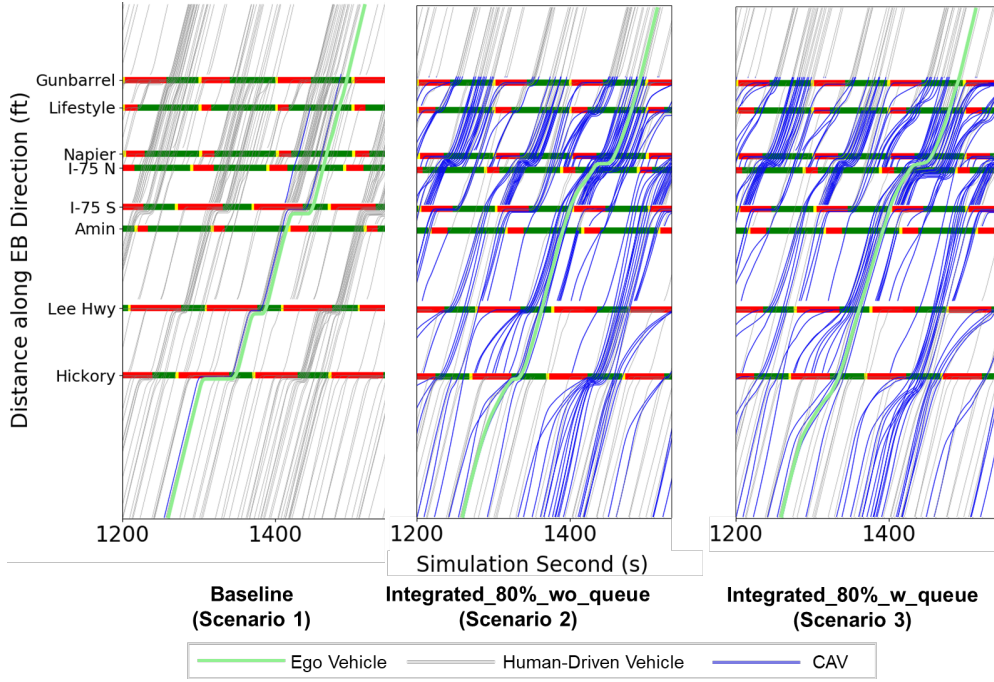


Fig. 5: The space-time diagram of three matched scenarios (80% MPR).

Figure 6 shows the VIL test results of battery consumption for the three matched scenarios mentioned previously. The ego vehicle's drive cycle (advisory speed profile) in Scenario 1 is generated by the VISSIM default driving behavior model, and the drive cycles in Scenarios 2 and 3 (integrated scenarios) are generated by the RAED speed planning algorithm with predicted bilinear signal controllers. Subfigure 6(a) indicates that the vehicle has better reference speed tracking performance in the integrated scenarios. In Scenario 1, the advisory speed profile increases and decreases the speed quickly. Therefore, the vehicle on the dynamometer is unable to follow the advisory speed profile very well. Because the vehicle runs in EV-only mode during dyno tests and its engine is off during the tests, only battery power contributes to the energy consumption. As shown in subfigure 6(b), the changing pattern of the battery power aligns with the need to increase or decrease the vehicle speed. Large positive battery power peaks correspond to the areas of large acceleration. Negative battery power occurs when regenerative braking occurs. In Scenario 1, the battery power fluctuates more significantly and has higher battery power peaks compared with that of Scenarios 2 and 3 because of rapid changes in vehicle speed. The figure shows that energy savings for Scenarios 2 and 3 are 26.3% and 26.9%, respectively, compared with that of the baseline Scenario 1. These energy benefits resulting from VIL testing are quite close to the simulation results, where the simulated energy benefits for these two corresponding comparisons are 27.4% and 28.7%, respectively. In addition to this group of scenarios, the testing results of the other three groups of scenarios (nine drive cycles) provide

477 similar findings, which implies that the VISSIM simulation can provide relatively
 478 representative evaluation results for the energy efficiency research of CAVs.

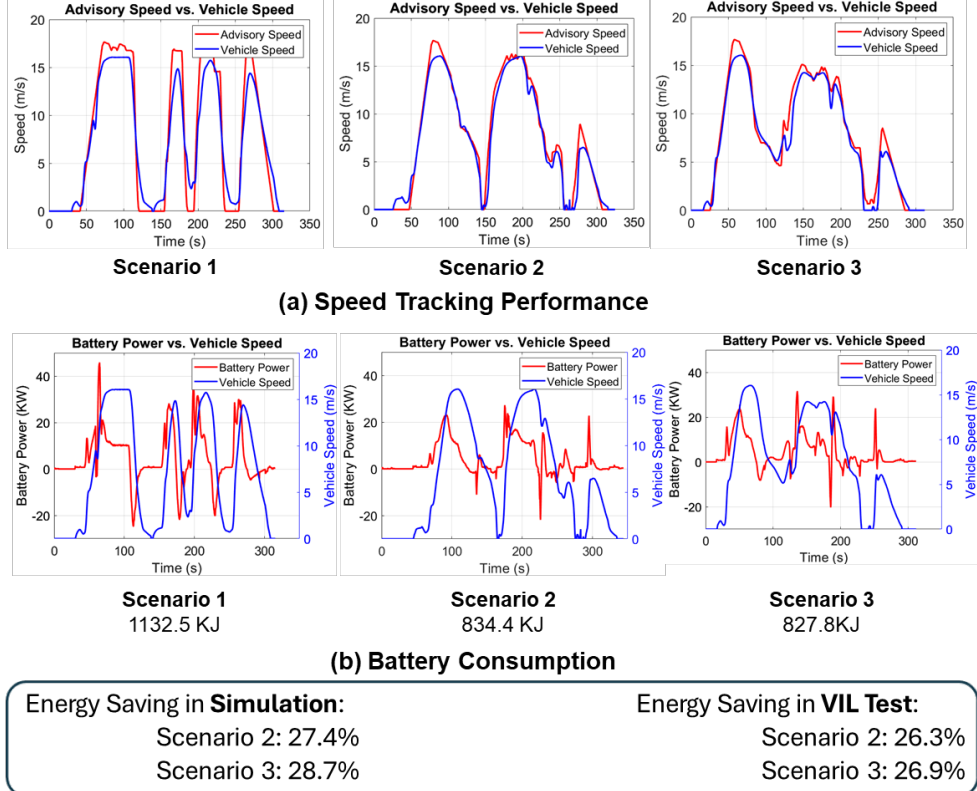


Fig. 6: The vehicle speed tracking performance and actual battery consumption of three matched scenarios (80% MPR).

479 The tracking behavior observed in Scenario 1 reflects limitations of the baseline
 480 advisory speed profile rather than limitations of the low-level speed-tracking MPC
 481 controller. In Scenario 1, the default VISSIM driving behavior model often produces
 482 abrupt and non-smooth acceleration and deceleration patterns that are difficult for
 483 the MPC controller to track and are not representative of physically realistic vehicle
 484 motion. In contrast, Scenarios 2 and 3 both use the RAED high-level speed planning
 485 algorithm, which produces smoother and more anticipatory trajectories that naturally
 486 lead to better tracking performance.

487 The distinction between Scenarios 2 and 3 highlights the role of queue prediction
 488 within the RAED framework. Scenario 2 applies RAED without queue prediction,
 489 while Scenario 3 includes real-time queue prediction. Because both use RAED, their
 490 tracking quality is similar, but the benefit of queue prediction is captured in the struc-
 491 ture of the advisory trajectory itself. In Scenario 3, the additional information on

492 anticipated downstream queue length allows RAED to initiate deceleration earlier,
493 avoid unnecessary stops, and better align the ego vehicle with upcoming green win-
494 dows. These effects can be seen in the smoother speed profiles and reduced number
495 of stops in Scenario 3 when compared with Scenario 2. Therefore, the primary impact
496 of queue prediction is on the quality of the advisory speed profile rather than on the
497 tracking accuracy of the low-level MPC controller.

498 It is worth mentioning that, the roadway slope and other environmental factors
499 are intentionally excluded from the VIL experiments. The dynamometer configuration
500 assumes a flat roadway in order to isolate the effects of the control algorithms on
501 vehicle behavior and energy consumption under repeatable conditions. The influence
502 of roadway elevation and environmental variability is instead captured in the on-road
503 field experiments described in Section 2.5, where the vehicle operates on the actual
504 roadway with real elevation profiles and uncontrolled traffic conditions.

505 On-Road Field Demonstration and Evaluation

506 To demonstrate the feasibility of the proposed integrated traffic-vehicle control system
507 under real-world conditions, a series of on-road experimental tests were conducted on
508 Shallowford Road in Chattanooga, Tennessee, with all system components fully inte-
509 grated, as illustrated in Fig. 7(a). The vehicle-side implementation uses two onboard
510 computing units, referred to as the Mapping PC and the Control PC, which are con-
511 nected through a shared onboard router using bidirectional UDP communication. The
512 Mapping PC interfaces with the global navigation satellite system (GNSS) receiver and
513 the cloud communication network to obtain vehicle localization, distance-to-stop-bar
514 information, SPaT data, and estimated queue states, and it executes the RAED high-
515 level speed planning algorithm to generate the advisory speed profile. The Control PC
516 interfaces directly with the forward-facing radar sensor and the vehicle controller area
517 network via the dSPACE MicroAutoBox II, executes the low-level MPC module to
518 enforce safe car-following and collision-avoidance constraints, and implements throttle
519 and braking commands. Radar-derived measurements, including lead-vehicle distance
520 and relative speed, are transmitted from the Control PC to the Mapping PC, while
521 advisory speed profiles are transmitted from the Mapping PC to the Control PC for
522 real-time execution. On the infrastructure side, the cloud-based RTMCS collects SPaT
523 and traffic flow data from GridSMART camera sensors and from an ORNL server con-
524 nected to the local traffic signal controllers through a virtual machine managed by the
525 City of Chattanooga. The ORNL server parses real-time NTCIP data and forwards
526 decoded SPaT information to the RTMCS, enabling closed-loop coordination between
527 traffic signal control and CAV speed planning during the field experiments.

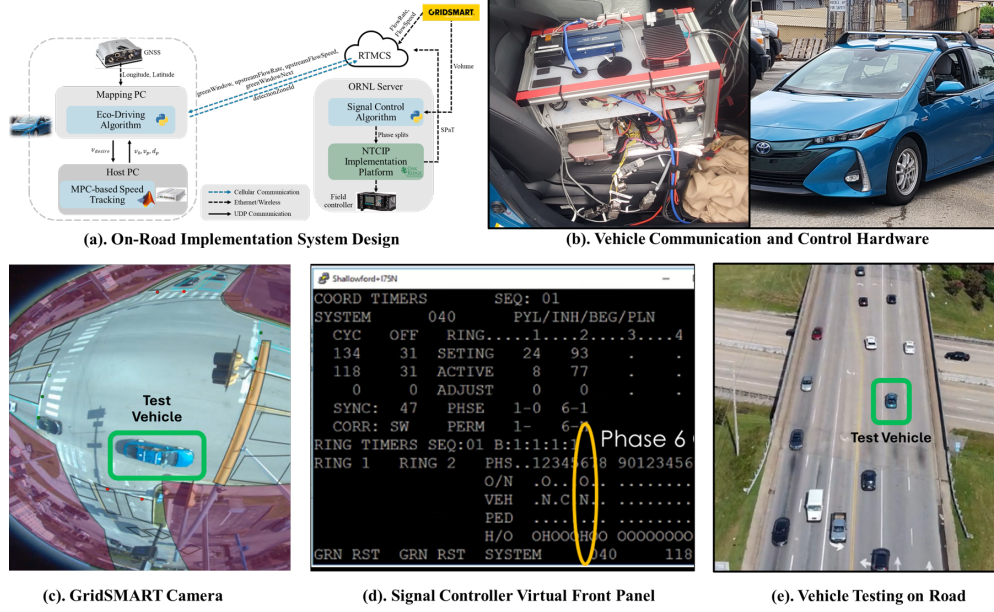


Fig. 7: Integrated traffic-vehicle control system field evaluation. (a) The implementation system design, (b) vehicle communication and control hardware, (c) the view of GridSMART camera, (d) signal controller virtual front panel, and (e) the drone view of vehicle testing on road.

Figure 7(b) shows the in-vehicle communication and control hardware, as well as the 2017 Prius Prime Plug-in Hybrid vehicle, which operated in EV-only mode during the experiment. Figure 7(c) is a screenshot from the GridSMART camera at the Lifestyle Way intersection, which captured the test vehicle during the experiment. Figure 7(d) shows the virtual monitor panel of the local traffic signal controller at the I-75 northbound intersection, which displays all the dynamic statuses of the controller. Figure 7(e) is captured from a drone video recorded during the experiment, showing the test vehicle in the middle of the image.

This section aims to evaluate and compare the effectiveness of different traffic signal control strategies with an on-road field experiment, specifically focusing on baseline actuated traffic signal control and the proposed bilinear traffic signal control. The key performance metric used in this comparison is the pattern of vehicle arrivals in relation to traffic signal phases—particularly the proportion of vehicles arriving during the green or red phases, also known as arrival on green (AOG) or arrival on red (AOR). AOG is a crucial indicator in traffic engineering because it reflects the efficiency of a traffic signal control system in minimizing delays and improving traffic flow. A higher rate of vehicle arrivals during the green phase suggests that the traffic signal control strategy is effectively synchronized with traffic demand, thereby reducing **stoppage time and fuel consumption**. In contrast, a higher proportion of AOR indicates potential areas for improvement in signal timing and coordination. AOG and AOR

denote the specific time steps at which a vehicle reaches the target intersection, usually measured by the stop bar detector, as shown in Figure 8. The stop bar detector, though critical for measuring traffic flow, encounters certain limitations when used to identify arrival patterns. For instance, it is inherently designed to detect vehicles as they pass; therefore, it may identify the arrival pattern of only the first vehicle that comes to a halt at the stop bar. This characteristic can prevent the detection of subsequent vehicles arriving on red, particularly in situations in which queueing occurs.

To address these challenges, we introduced a dynamic arrival time estimation (DATE) approximation method to infer the precise timing of vehicle arrivals at the target intersection, as shown in Figure 8(a). Although queues may extend upstream, it is often difficult for them to reach the outgoing detector of the upstream intersection. Therefore, the activation time recorded by the outgoing detector at the upstream intersection is considered a reliable indicator of vehicle departure. This activation time is synchronized with the signal phase at the target intersection to distinguish whether the vehicle has started its journey on a green signal (start on green [SOG]) or on a red signal (start on red [SOR]). The estimated travel time is calculated using the vehicle travel speed detected by the outgoing detector, v_{out} , and the distance between the outgoing detector and the stop bar detector, $d_{out2stop}$. This estimated travel time, t_{est} , is then added to the departure time recorded by the outgoing detector, providing the estimated arrival time at the target intersection. By comparing the estimated travel time t_{est} to the remaining duration of the green-light phase of the target intersection and direction, the team can determine whether a vehicle will arrive at the intersection during the green or red light. This methodology overcomes the limitations of stop bar detection in heavy traffic conditions and provides a more comprehensive understanding of traffic flow, enabling a detailed assessment of traffic signal control strategies.

Figure 8(b) presents the calculated probabilities of AOG and AOR under different initial conditions. These conditions are segmented according to whether the vehicle starts during a green (SOG) or red (SOR) signal at the outgoing detector at the upstream intersection. The results indicate a superior performance of the bilinear controller compared with that of the actuated controller—as evidenced by the higher AOG rates of 84.67% and 34.65%, respectively—regardless of the signal status at the start time. This suggests that the bilinear control system is more effective at accommodating a greater number of vehicles and increasing overall throughput. Furthermore, the results indicate a substantially higher likelihood of AOR when vehicles begin their trips during a red signal phase, with AOR rates of 75.64% and 68.36%, compared with 19.68% and 15.33% when trips begin during a green phase, which is consistent for both control systems. This outcome may imply that the distances between adjacent intersections are relatively short, necessitating an improvement in coordination strategies to minimize red light arrivals and optimize traffic movement.

Figure 8(c) illustrates the comparative analysis of time to green (TTG) for vehicles that are estimated to be arriving on red, comparing the performance of actuated and bilinear traffic control systems. TTG is quantified as the interval between a vehicle’s arrival at a red signal and the start of the subsequent green phase, essentially measuring the vehicle’s expected waiting time at the intersection. Under actuated control, the distribution of TTG is observed to shift toward longer waiting times compared with

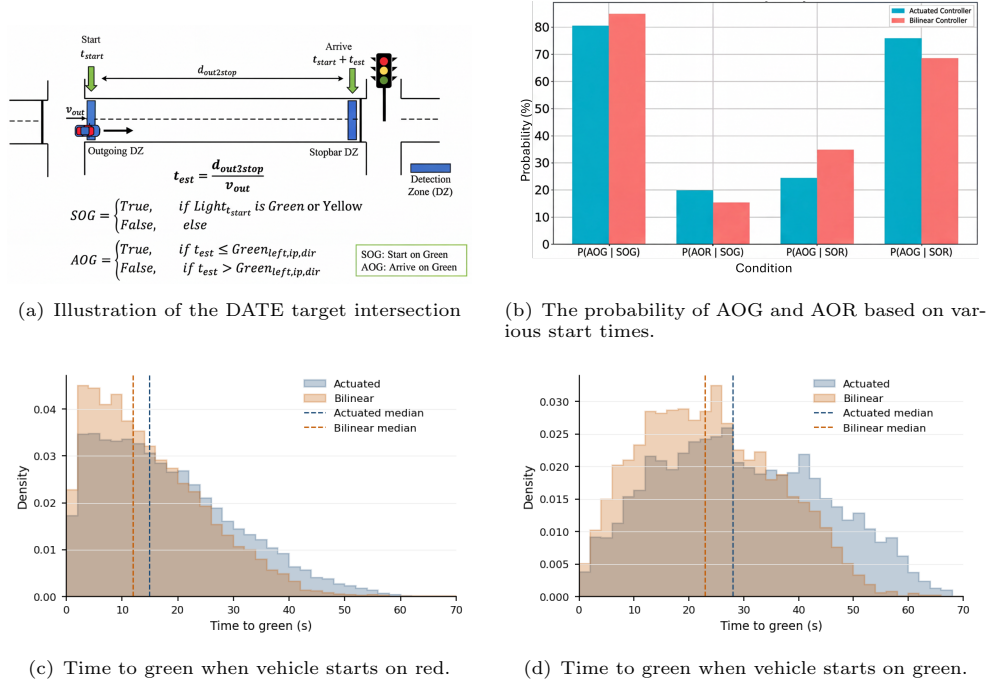


Fig. 8: DATE Performance Evaluation.

those of bilinear control. This shift suggests that vehicles may experience increased delays before the start of green when actuated control is implemented. When comparing SOR and SOG scenarios, SOR vehicles show a steeper decline in frequency as TTG increases, indicating that vehicles are less likely to encounter long waiting times at the intersection when they begin their trips during a red signal phase. The higher arrival-on-green rates observed under predictive bilinear control demonstrate the benefit of multi-cycle signal optimization, which enhances progression and reduces stop-and-go behavior in the field.

Figure 9 shows the time series data from two intersection-approaching scenarios in the field demonstration, including speed, acceleration, eco-driving enable status, battery power, roadway elevation, cumulative energy consumption, and distance to the stop bar. Both scenarios are from the test drive on the eastbound side of the Lifestyle Way and Shallowford Road intersection. Figure 9(a) (Scenario 1) shows data gathered without eco-driving, whereas Figure 9(b) (Scenario 2) shows data with eco-driving enabled. The green and red bars at the bottom of the figure represent the signal light status. In both scenarios, the test vehicle starts on a red light and approaches the downstream traffic light. The signal light turns green when the vehicle is about to arrive at the stop bar (about 400 ft upstream). With eco-driving enabled, the vehicle starts to decelerate before getting close to the stop bar, at which time the regenerative braking (i.e., negative battery power) starts to recapture the kinetic energy. Conversely,

in the scenario without eco-driving enabled, the vehicle brakes primarily when arriving at the intersection or the end of the queuing vehicles, which will consume more energy. When the test vehicle reaches the intersection, enabling eco-driving results in a significant energy savings of approximately 23%, reducing consumption from 145 kJ to 112 kJ. However, the real-world traffic environment is not controlled between these two scenarios. Specifically, the ego vehicle in Scenario 1 encountered a queue (about 250 ft) when approaching the stop bar, whereas the ego vehicle in Scenario 2 did not encounter any stopped vehicles during its approach. The difference in the downstream queuing status would also affect the energy efficiency of the ego vehicle.

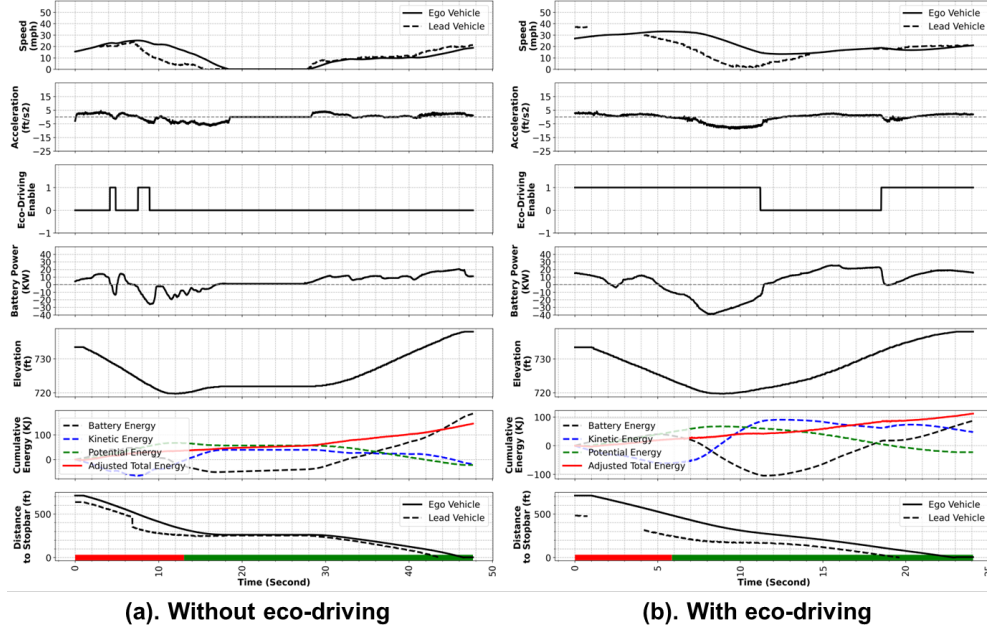


Fig. 9: Comparisons of time to green for vehicles arriving on red.

As summarized in Figure 2, performing an exact apples-to-apples energy evaluation in a real-world traffic environment is challenging. Too many energy influence factors come into play, including traffic flow characteristics and relative TTG/red at departure, which are difficult to control during the on-road experiment. Therefore, the previous example of 23% energy savings is only to demonstrate the feasibility and capability of energy savings potential rather than to provide a statistically solid conclusion. During the field demonstration, 35 scenarios in 7 intersection approaches were collected, among which 25 were eco-driving enabled and the remaining 10 were eco-driving disabled.

Figure 10 presents cumulative energy consumption by the distance to the stop bar for all the scenarios at four intersection approaches that have relatively good sample sizes. The dashed red line represents the mean value of the cumulative energy

consumption for all the baseline scenarios (eco-driving disabled) at the corresponding intersection approach. Similarly, the dashed green line represents the mean value of the cumulative energy consumption for all the eco-driving enabled scenarios. The shaded regions represent the standard deviation of cumulative energy consumption across repeated test runs for each scenario. The absence of red shading in subplot (d) indicates that only a single baseline sample was available for that intersection approach, and therefore no standard deviation could be computed. The figure shows that when the test vehicle reaches the intersection (i.e., when the distance to the stop bar is 0 ft), enabling eco-driving results in an average energy savings of approximately 49%, 29%, 57%, and 3% for the four intersection approaches, respectively.

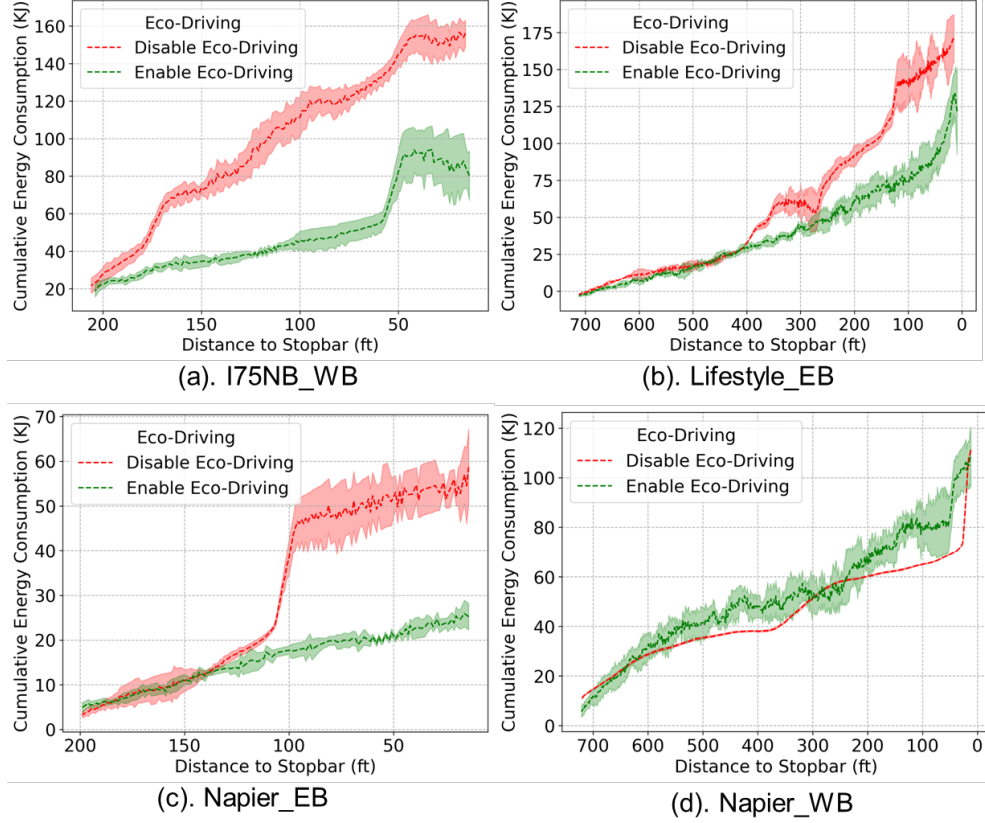


Fig. 10: Cumulative vehicle energy consumption as a function of distance to the stop bar under baseline and eco-driving conditions.

Unlike the simulation environment, a field demonstration can hardly provide a statistically significant apples-to-apples comparison because the number of tested scenarios is quite limited, and many energy-sensitive factors are not controlled during the

experiment (e.g., relative signal status and surrounding traffic environment). In addition, different intersections have their own characteristics that may affect the energy benefit of the eco-driving algorithm (e.g., roadway elevation, segment length, and signal timing characteristics). As shown in Figure 10, the energy savings results are quite diverse among different intersection approaches, ranging from 3% to 57%. Although field experiments demonstrate that eco-driving control can save a large amount of energy, the absolute percentage of savings is not considered a very convincing number owing to the small sample size and uncontrolled experimental factors.

The proposed framework is real-time implementable at both the infrastructure and vehicle layers, but at different control time scales. At the infrastructure layer, the predictive bilinear signal controller operates at the signal-cycle level. The bilinear model parameters, i.e., matrices A , B , and C in Eq. (2), are updated online once per signal cycle using a normalized least-squares algorithm, whose computational cost scales primarily with the number of traffic movements and controllable phases. The signal timing optimization in Eqs. (4) and (8) is then solved cycle-by-cycle using the conjugate gradient method implemented in SciPy. In the field corridor, all eight signalized intersections operate with a fixed 100-second cycle length, which provides sufficient time for online optimization and generation of multi-cycle SPaT forecasts before the next control update. At the vehicle layer, the RAED planner and low-level MPC controller are executed onboard the test vehicle. The Mapping PC receives GNSS, distance-to-stop-bar, SPaT, and queue-state information and updates the RAED advisory speed at a 1-second planning step. The Control PC receives the advisory speed profile, radar-based lead-vehicle information, and vehicle CAN signals, and executes the MPC-based longitudinal controller through the dSPACE MicroAutoBox II interface. Thus, the signal controller was implemented in real time at the cycle level, while the RAED/MPC vehicle controller was implemented in real time at the onboard planning and longitudinal-control level.

Methods

This section describes the algorithms and system components that constitute the integrated traffic-vehicle control system evaluated in Section 2. Building on the system overview in Fig. 1, the methods are organized according to the two interacting subsystems of the framework. First, the traffic signal control subsystem is presented, which uses predictive bilinear optimization to update signal timings over multiple future cycles based on real-time traffic information. Second, the vehicle control subsystem is described, which focuses on longitudinal CAV speed planning using the RAED algorithm, with queue prediction incorporated as part of the high-level planning process. A low-level model predictive control module is used solely to execute the planned speed trajectories under real vehicle dynamics. Finally, the cyber-physical infrastructure that enables real-time communication between vehicles and traffic signals is introduced. Together, these components form a closed-loop system that coordinates traffic signal control and CAV speed planning to achieve the performance improvements reported in the Results section.

Predictive Bilinear Control for Traffic Signals

The urban traffic signal control problem aims to calculate optimal traffic signal timings cycle-by-cycle to reduce traffic delays. In this study, the team extends the previous study of bilinear traffic signal control models where real-time detector or connected-vehicle data are used to update signal timing decisions in response to traffic conditions [47]. Wang et al.[47] provides the direct methodological foundation for the bilinear traffic-flow model, online parameter-estimation procedure, and single-cycle signal optimization formulation used in this study. The present work extends that formulation from a single-cycle adaptive controller to a multi-cycle predictive controller by iteratively propagating traffic states and optimizing phase splits over future cycles. This predictive extension allows the signal controller to provide forward-looking SPaT forecasts to the vehicle-side planner, which is necessary for integrated signal-vehicle coordination. Moreover, by modifying the objective function in predictive bilinear control from minimizing travel delay to maximizing traffic volume (throughput), the network performance (delay and energy consumption) is improved. This section summarizes the technical details of the bilinear control model and describes how the project team extends the model in this study.

A discrete-time multi-input and multi-output model is applied for traffic system modeling, as shown in Equation (1):

$$z(k+1) = F(z(k), v(k)), \quad (1)$$

where F denotes the nonlinear relationships between traffic volumes z and phase duration v . The k denotes time step (a cycle with fixed time, e.g., 100 s).

As described in the previous study [47], the system in Equation (1) can be approximated by a bilinearized state-space form, as shown in Equation (2):

$$\Delta z(k+1) = A\Delta z(k) + B\Delta v(k) + (C\Delta v(k)) \odot \Delta z(k), \quad (2)$$

where $A \in R^{(M \times M)}$, $B \in R^{(M \times N)}$, and $C \in R^{(M \times N)}$ are the system parameters that can be estimated online (cycle-by-cycle) using a normalized least squared algorithm. The details are omitted here because the previous study [47] documents the step-by-step parameter update law. The $\Delta z(k)$ and $\Delta v(k)$ are the volume increment and phase duration increment at k th time step, and the operator $\odot$ is the elementwise multiplication. Equation (2) can be reformulated as Equation (3) if $y(k) = \Delta z(k)$ and $u(k) = \Delta v(k)$ are defined.

$$y(k+1) = Ay(k) + Bu(k) + (Cu(k)) \odot y(k) \quad (3)$$

Note that $y(k+1) = \Delta z(k+1) = z(k+1) - z(k)$ is the traffic volume increment. Because the volume at current time step $z(k)$ is known (e.g., collect from traffic detectors at k), maximizing the future volume $z(k+1)$ is equivalent to maximizing the volume change $y(k+1)$. Hence, the objective of bilinear control is to maximize the volume change of all movements M at $k+1$ by computing the optimal signal timings $v^*(k+1)$.

$$v^*(k+1) = \arg \max_{v(k+1)} \sum_{m=1}^M [y_m(k+1)] \quad (4)$$

The optimal signal timings for all movements are bounded by the maximum and minimum green splits, as shown in Equation (5):

$$s.t. v_{min} \leq v^*(k+1) \leq v_{max} \quad (5)$$

Another constraint is the fixed cycle length constraint defined in Equation (6). For each intersection, all phases in one ring sum up to a fixed value of C (e.g., $C = 100$ s).

$$\sum_{p=r}^R v_p = C, \quad (6)$$

where r denotes the ID of the first phase in a ring and R denotes the ID of the last phase in a ring in a Ring Barrier Controller [52].

Equations (1) through (6) demonstrate the use of a bilinear control model to compute optimal signal timings for the next cycle based on current traffic conditions. To plan CAV trajectories and reduce energy consumption, optimal signal timing information for upcoming cycles must be provided in advance. Therefore, a predictive bilinear control model has been developed. Equation (7) predicts traffic volume for cycle $k+i$ ($i \geq 2$), and Equation (8) aims to maximize the traffic volume for all movements M in future cycle i based on the predicted volume from (7). For computing signal timings for additional future cycles ($i \geq 2$), Equations (7) and (8) are applied iteratively to generate optimal signal timings $v^*(k+i)$. The optimization problems in Equations (4) and (8), along with the online parameter estimation, can be solved using the conjugate gradient method implemented in SciPy [53].

$$\hat{y}(k+i) = A\hat{y}(k+i-1) + B(v(k+i) - v^*(k+i-1)) + C(v(k+i) - v^*(k+i-1)) \odot \hat{y}(k+i-1) \quad (7)$$

$$v^*(k+1) = \arg \max_{v(k+1)} \sum_{m=1}^M [\hat{y}_m(k+i)] \quad (8)$$

The predictive extension does not change the objective function itself, but applies it iteratively over multiple future cycles using predicted traffic states, which enables coordination with downstream vehicle speed planning.

This is a real-time framework and contains the learning of the system as denoted in the online estimation of the parameter matrices A , B , and C using the online least squares algorithm. Because the system is of a bilinear structure in which parameter matrices are linearly related to the system output $y(k)$, the parameter estimation in the closed-loop system should converge. In this context, once all the eigenvalues of

the estimated matrix A are inside a unit circle of the complex plan, the closed-loop system should always be stable once the optimized signal timing plan v^* is uniformly bounded. This is a well-known result from an adaptive control system perspective.

Compared with existing actuated or reactive controllers, the predictive extension improves progression by optimizing signal timing over multiple future cycles, enabling higher AOG rates as demonstrated in Section 2.4.

Vehicle Control System

The vehicle control system consists of two levels: (1) a high-level speed planning module that generates an energy-efficient and socially compatible advisory speed profile, and (2) a low-level MPC that ensures accurate longitudinal execution. This hierarchical design enables real-time adaptation to downstream SPaT, predicted queue dynamics, and interactions with surrounding human-driven vehicles (HDVs), while maintaining smooth and safe vehicle motion in mixed traffic.

High-level speed planning: Roadmanship-aware eco-driving (RAED)

The RAED planner is related to prior eco-approach/departure and speed-planning methods for CAVs at signalized intersections, in which vehicle trajectories are optimized using SPaT information to reduce fuel or energy consumption [30]-[34]. Several of these studies also consider queue effects, multi-intersection corridors, or hierarchical speed tracking. Recent integrated signal-vehicle coordination studies further demonstrate that coupling traffic signal control with vehicle trajectory planning can improve mobility and energy performance in mixed traffic [35]-[39]. Building on these foundations, RAED is designed as an implementable onboard planner rather than a centralized trajectory optimizer. It uses predicted SPaT windows, field-accessible queue estimates, and ego-vehicle state information to generate advisory speeds in real time. Its main algorithmic distinction is the addition of a roadmanship-aware minimum-speed constraint, which prevents the planner from selecting excessively slow upstream coasting trajectories that may be energy-efficient for the ego vehicle but disruptive in mixed human-driven traffic.

The RAED algorithm [48] generates an advisory speed by combining ego-vehicle states, lead-vehicle behavior, predicted queue lengths, and current/predicted SPaT information. A key physical element is the vehicle-specific coasting model.

$$a_{\text{coast}} = -\frac{f_0 + f_1 v + f_2 v^2}{M}, \quad (9)$$

where f_0, f_1, f_2 are the road-load (coastdown) coefficients, M is vehicle mass, and v is current speed. For the 2017 Toyota Prius Prime used in this study [48]: $f_0 = 0.083698 \text{ kW}\cdot\text{s/m}$, $f_1 = 0.00385 \text{ kW}\cdot\text{s}^2/\text{m}^2$, $f_2 = 0.000278 \text{ kW}\cdot\text{s}^3/\text{m}^3$, and $M = 1.6443 \text{ metric tons}$.

792 To ensure arrival during feasible downstream green windows, RAED computes
 793 allowable cruise velocities. For the first downstream intersection,

$$C_{11} = \left[\frac{d_1}{t_{12} - t_{\text{curr}}}, \frac{d_1}{t_{11} - t_{\text{curr}}} \right] \cap [0, v_{\text{max}}], \quad (10)$$

794 where d_1 is distance to the stop bar, $[t_{11}, t_{12}]$ is the next green interval, and v_{max} is
 795 the speed limit. Similar expressions generate C_{12}, C_{21}, C_{22} . The feasible set is

$$C = (C_{11} \cap C_{21}) \cup (C_{12} \cap C_{22}). \quad (11)$$

796 To maintain socially compatible behavior with surrounding HDVs, RAED enforces
 797 a roadmanship constraint

$$v_k \geq v_{\min}(d_k), \quad (12)$$

798 where d_k is distance to the intersection and $v_{\min}(d_k)$ is a distance-dependent minimum
 799 speed. This roadmanship constraint serves as a behavioral regularization mechanism
 800 for mixed-traffic operation. Without this constraint, a purely energy-optimal eco-
 801 driving planner may select an excessively low advisory speed far upstream of the
 802 intersection to maximize coasting distance and align the vehicle precisely with the start
 803 of a downstream green window. Although this behavior may reduce ego-vehicle energy
 804 consumption, it can be impractical in human-driven traffic because it may create large
 805 speed differentials relative to surrounding vehicles, increase headway variability, and
 806 induce unnecessary braking by following drivers. The distance-dependent formulation
 807 of $v_{\min}(d_k)$ prevents unrealistically slow upstream coasting while still allowing grad-
 808 ual deceleration near the stop bar, where lower speeds may be necessary because of
 809 queues or red-light compliance. Thus, the constraint balances energy-efficient coasting
 810 with traffic-flow compatibility and socially acceptable longitudinal behavior.

811 Specifically, $v_{\min}(d_k)$ is implemented as a piecewise step function parameterized
 812 by two distance thresholds and two speed-limit ratios:

$$v_{\min}(d_k) = \begin{cases} \alpha_1 v_{\text{lim}}, & d_k \geq D_1, \\ \alpha_2 v_{\text{lim}}, & D_2 \leq d_k < D_1, \\ 0, & d_k < D_2, \end{cases} \quad (13)$$

813 where $v_{\text{lim}} = 40$ mph (17.88 m/s) is the posted corridor speed limit, $D_1 = 700$ m,
 814 $D_2 = 400$ m, $\alpha_1 = 0.75$, and $\alpha_2 = 0.50$. Far from the intersection ($d_k \geq D_1$), the
 815 vehicle is required to maintain at least 75% of the speed limit to avoid impeding
 816 surrounding traffic. Within 400–700 m, the lower bound relaxes to 50% of the speed
 817 limit. Within 400 m of the stop bar, no lower bound is enforced, permitting the vehicle
 818 to decelerate fully for a red-light stop or upstream queue.

819 Finally, RAED selects the advisory eco-driving speed using a coasting-based heuris-
 820 tic. Let $s_{\text{coast}}(v)$ denote the predicted coasting distance when starting at speed v . The
 821 chosen advisory speed is

$$v^* = \arg \min_{v \in C} |s_{\text{coast}}(v) - d_1|, \quad (14)$$

822 which yields the speed whose natural coasting trajectory aligns best with the
 823 downstream intersection and green timing.

824 The roadmanship-aware formulation differentiates RAED from prior eco-driving
 825 algorithms by explicitly incorporating constraints that ensure socially compatible
 826 behavior in mixed traffic, a feature validated in both VIL and on-road experiments.

827 Low-level speed tracking: Model Predictive Control

828 To execute the advisory speed profile generated by RAED, we employ a low-level lon-
 829 gitudinal controller based on MPC. MPC is well suited for this application because
 830 it explicitly handles the competing requirements of accurate speed tracking, passen-
 831 ger comfort, actuator limits, and collision avoidance. At each control step, the MPC
 832 module predicts the vehicle's future motion using a discrete-time kinematic model and
 833 computes optimal acceleration and deceleration commands that minimize deviations
 834 from the RAED advisory speed while respecting comfort and safety constraints. The
 835 prediction model of the MPC controller is shown in Equations (15) and (16):

$$x_{ego}(k+1) = x_{ego}(k) + v_{ego}kT \quad (15)$$

$$v_{ego}(k+1) = v_{ego}(k) + a_{ego}kT, \quad (16)$$

836 where x_{ego} is the ego vehicle position, v_{ego} is the ego vehicle velocity, a_{ego} is the ego
 837 vehicle acceleration, k is the current sampling time, $k+1$ is the next sampling time,
 838 and T is the total sampling time period. The state-space form of the model is shown in
 839 Equations (17) and (18). The state vector is defined as $X = [x_{ego} \ v_{ego}]^T$. The control
 840 input is defined as $u = a_{ego}$.

$$X(K+1) = AX(K) + BU(K) \quad (17)$$

$$A = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ T \end{bmatrix}. \quad (18)$$

841

842

843 The ego vehicle jerk, j_{ego} , can be calculated with Equation (19):

$$\dot{j}_{ego}(k) = \frac{a_{ego}(k) - a_{ego}(k-1)}{T}. \quad (19)$$

844

845 The objective of the MPC controller is to find an optimal control vector,
 846 $a_{ego}(k)$ in Equation (20), in the control horizon that will minimize the cost, J in
 847 Equations (21) and (22).

$$a_{ego}(k) = [a_{ego}(1|k) \ a_{ego}(2|k) \ \dots \ a_{ego}(N-1|k)]^T \quad (20)$$

$$a_{ego}^* = \arg \min_{a_{ego}} J, \quad (21)$$

$$J = \sum_{i=k}^{k+N-1} \left(q_{\text{accel}} a_{\text{ego}}^2(i|k) + q_{\text{jerk}} j_{\text{ego}}^2(i|k) + q_{\text{advisory}} (v_{\text{ego}}(i|k) - v_{\text{ref}}(i|k))^2 \right) + q_{\text{track}} (x_{\text{ego}}(N|k) - x_{\text{track}}(N|k))^2, \quad (22)$$

where q_{accel} , q_{jerk} , q_{advisory} , and q_{track} are cost weights, v_{ref} is the desired reference speed profile, and x_{track} is the desired ego vehicle position at the end of the prediction horizon. The cost function consists of four terms. The first two terms, which minimize the acceleration and jerk of the ego vehicle, are designed for driver comfort. The third term controls the ego vehicle speed to track the reference speed. The fourth term keeps the position of the ego vehicle at the end of the prediction horizon at $x_{\text{track}}(N|k)$, which is the minimum distance to a stop line and the position constraint to avoid hitting a lead vehicle. The constraints of the optimization problem are defined in Equations (23) through (26),

$$v_{\min} \leq v_{\text{ego}} \leq v_{\max} \quad (23)$$

$$x_{\text{ego}} \leq x_{\max} \quad (24)$$

$$a_{\min} \leq a_{\text{ego}} \leq a_{\max} \quad (25)$$

$$j_{\min} \leq j_{\text{ego}} \leq j_{\max}, \quad (26)$$

where $v_{\min}$ and $v_{\max}$ are the speed limits, $x_{\max}$ is the maximum position constraint in the horizon, $a_{\min}$ and $a_{\max}$ are acceleration limits, and $j_{\min}$ and $j_{\max}$ are jerk limits. The optimization is solved using a Newton-GMRES method with Fischer-Burmeister complementarity smoothing [54], executing 10 Newton iterations with 20 GMRES iterations per Newton step.

The MPC implementation uses the following parameters: prediction horizon $N = 20$ steps at a step size of $\Delta t = 0.5$ s (10-second lookahead), executed at 5 Hz ($\Delta t_{\text{exe}} = 0.2$ s). Constraint bounds are $v \in [0, 22.4]$ m/s, $a \in [-3, 2]$ m/s², and rate of acceleration $\in [-1, 1]$ m/s³. The four cost weights are $q_{\text{accel}} = 0.1$, $q_{\text{jerk}} = 1.0$, $q_{\text{advisory}} = 0.1$, and $q_{\text{track}} = 0.05$ (applied as a terminal penalty). Each solve completes in approximately 50–70 ms, using 25–35% of the 200 ms execution window.

Cyber-Physical Infrastructure

To support the real-time and scalable deployment of the integrated traffic-vehicle control system described in earlier sections, the team developed and employed a cloud-based digital twin platform as the core cyberinfrastructure [55–57]. This platform facilitates the dynamic integration of transportation infrastructure data, traffic control strategies, and V2I communication protocols across diverse experimental settings. As a result, the experimental framework becomes more adaptable and scalable, allowing deployment in varied urban environments with minimal infrastructure reconfiguration.

Table 1: Implementation parameters for the MPC and RAED controllers (2017 Toyota Prius Prime, Shallowford Road corridor).

Module	Parameter	Value	Unit
MPC	Horizon N	20	steps
	Step size Δt	0.5	s
	Execution rate	5	Hz
	$[v_{\min}, v_{\max}]$	$[0, 22.4]$	m/s
	$[a_{\min}, a_{\max}]$	$[-3, 2]$	m/s ²
	$[j_{\min}, j_{\max}]$	$[-1, 1]$	m/s ³
	$(q_{\text{accel}}, q_{\text{jerk}}, q_{\text{advisory}}, q_{\text{track}})$	(0.1, 1.0, 0.1, 0.05)	—
RAED	Solve time	50–70	ms
	f_0 (road-load coefficient)	0.083698	kW·s/m
	f_1 (road-load coefficient)	0.00385	kW·s ² /m ²
	f_2 (road-load coefficient)	0.000278	kW·s ³ /m ³
	M (vehicle mass)	1.6443	metric tons
	D_1 (speed threshold 1)	700	m
	D_2 (speed threshold 2)	400	m
	(α_1, α_2) (speed-limit ratios)	(0.75, 0.50)	—
	Advisory update period	1	s

877 The field implementation includes safety-oriented communication-handling
 878 logic for temporary interruptions in V2I data exchange. During short-term delays or
 879 missed SPaT updates, the vehicle-side module uses the most recently received valid
 880 SPaT message and locally propagates the remaining time of the current signal phase to
 881 avoid abrupt changes in the advisory speed. GPS-based map matching is used to main-
 882 tain localization consistency. If the received SPaT or queue information becomes stale
 883 or inconsistent for an extended period, the RAED advisory is suspended and the driver
 884 is notified, while the low-level vehicle controller and safety driver continue to maintain
 885 safe vehicle operation. The RTMCS deployed in this study is the same platform charac-
 886 terized in Xu et al. [56], in which end-to-end V2I communication latency was measured
 887 across 2,002 communication events during a 20-mile urban drive in Knoxville, TN,
 888 under a 4G network (download 115–128 Mbps; upload 19.3 Mbps). All 2,002 measured
 889 latencies were at most 0.9 s, and the median measured latency was 0.5 s. The onboard
 890 speed optimization algorithm executes in under 0.05 s [56], providing adequate mar-
 891 gin within each 1-second advisory update cycle. These measurements characterize
 892 the RTMCS operating latency range under real urban field conditions but were col-
 893 lected under normal network operation. Controlled stress tests under imposed latency,
 894 packet-loss, or range degradation were not conducted in either study. Therefore, the
 895 on-road experiments reported in this work should be interpreted as demonstrating
 896 deployment feasibility under naturally occurring field communication conditions, not
 897 as a controlled validation of robustness to degraded V2I communication.

898 Safety Assurance of Research Experiments

899 Safety was a central requirement in the design and execution of all evaluation stages,
 900 including simulation, Vehicle-in-the-Loop (VIL) testing, and on-road field experiments.

901 The integrated control framework was deployed only after multiple layers of safety pro-
902 tection were implemented and verified. The safeguards cover vehicle-side automation,
903 infrastructure-side signal control, and institutional oversight.

904 The proposed framework automates only longitudinal vehicle control. All lat-
905 eral maneuvers such as steering, lane keeping, and lane changes remained under the
906 full supervision of a trained safety driver. A professional test engineer from Toyota
907 Motor North America, who has extensive experience in experimental vehicle eval-
908 uations, served as the primary safety driver during all on-road tests and retained
909 complete authority over throttle and braking at all times. Factory-installed safety sys-
910 tems, including Automatic Emergency Braking, traction control, and stability control,
911 remained active throughout testing. Before deployment on public roads, the control
912 algorithms were validated in the VIL laboratory, which provided a controlled envi-
913 ronment for assessing longitudinal responses and safety-related behaviors under real
914 vehicle dynamics.

915 The predictive bilinear traffic signal controller was designed to operate strictly
916 within the safety limits enforced by standard traffic signal control hardware. Minimum
917 green times, yellow and all-red intervals, pedestrian crossing durations, and right-of-
918 way protections were never modified by the algorithm. Only the phase splits (i.e.,
919 distribution of green time among phases) was adjusted. During field testing, two traffic
920 engineers from the City of Chattanooga monitored real-time traffic and signal oper-
921 ations to ensure that all control actions remained safe for the public and compatible
922 with expected traffic patterns.

923 All experiments were conducted under Oak Ridge National Laboratory’s formal
924 safety review and approval process. The research team completed required training
925 and scenario-based risk assessments to identify potential hazards and establish mitiga-
926 tion procedures before on-road implementation. The experimental plan was reviewed
927 and approved by laboratory management and safety officers, ensuring compliance with
928 institutional protocols and public safety regulations. Together, these measures ensured
929 that every stage of the study was performed under strict safety constraints, and they
930 provide a clear justification for the controlled and responsible execution of the field
931 trial.

932 It is important to distinguish between operational safety assurance for
933 conducting the experiments and quantitative safety-performance validation of the
934 proposed control framework. In this study, safety assurance was achieved through
935 a combination of experimental safeguards, institutional review, human supervision,
936 active factory safety systems, and algorithmic constraints on longitudinal motion.
937 However, the experiments were not designed as a comprehensive safety-performance
938 assessment. The safety claims in this study are limited to the controlled and supervised
939 execution of the experimental deployment, together with the enforcement of longitu-
940 dinal speed, position, acceleration, and jerk constraints in the MPC controller. A full
941 quantitative safety evaluation under diverse mixed-traffic conditions is left for future
942 work.

943 Discussions

944 This study proposed a framework for the development and evaluation of integrating
945 energy-efficient traffic signal control and CAV control and then validated the system
946 through a three-stage process: traffic simulations, VIL testing, and on-road demon-
947 strations. A detailed VISSIM simulation model was developed and calibrated based
948 on real-world data from the Shallowford Road corridor in Chattanooga, Tennessee.
949 In total, 16 scenarios (one baseline scenario with no vehicle control and 15 inte-
950 grated traffic-vehicle control scenarios with 5 different MPRs and 3 types of queue
951 effect considerations) were simulated and evaluated. A route-based pairwise (CAVs
952 vs. human-driven vehicles) energy evaluation was conducted. The experimental test
953 results indicate that an electrified CAV with the ideal queue length prediction can
954 achieve up to 15.92% energy savings in the eastbound direction, and the same scenario
955 can achieve up to 20.16% energy savings in the westbound direction.

956 The consideration of queue prediction can significantly increase the energy
957 savings achieved by the integrated traffic-vehicle control system, and the imple-
958 mentable queue prediction performs nearly as well as the ideal queue prediction. This
959 finding indicates that the queue length prediction algorithm should be integrated in
960 real-world deployment. It is important to emphasize that the ideal queue prediction
961 scenario represents an upper-bound reference rather than a practical deployment case.
962 The main conclusions of this study are therefore drawn from the implementable queue
963 prediction results, which rely on field-accessible sensing and communication technolo-
964 gies. The close performance gap observed between the ideal and implementable cases
965 suggests that substantial energy savings can be achieved without requiring perfect
966 knowledge of all vehicle states.

967 In addition to traffic simulations, four groups of drive cycles (i.e., 12 drive
968 cycles, each group having 3 matched comparison drive cycles) with different penetra-
969 tion rates, traffic signal control methods, and speed control algorithms were identified
970 from the traffic simulation results and have been tested on the VIL platform. The
971 VIL test results validate that the VISSIM simulation is able to provide reliable
972 energy-saving results for energy efficiency research.

973 Moreover, complete on-road testing that simultaneously controls the traffic
974 signals and CAV in the real-world has been conducted. **This study implemented and**
975 **demonstrated an integrated traffic-vehicle control system in real-world traffic scenar-**
976 **ios.** The on-road testing of the integrated control has provided valuable insights into
977 the practical application of integrated vehicle and traffic signal control technologies.
978 The tests demonstrated a clear potential for reduced vehicle energy consumption and
979 improved traffic efficiency under controlled conditions. The vehicle control system
980 effectively synchronized vehicle speeds with traffic signals, optimizing arrivals during
981 green phases and thus minimizing stops and delays. Specifically, in the integrated traf-
982 fic signal control and vehicle control testing, the project team found that the bilinear
983 traffic signal control itself generates benefits in improving the vehicle's AOG percent-
984 age (about 4%–10%), reducing the number of stops and improving traffic flow and
985 energy efficiency. Regarding the vehicle controller benefits, the project team found
986 that the energy savings are quite diverse among different intersection approaches,

ranging from 3% to 57%. Although the field experiments demonstrate that the eco-driving control can save a large amount of energy, the absolute percentage of savings is still not considered very convincing because of the small sample size and uncontrolled experimental factors.

It is also worth noting that the control framework developed in this study is limited to longitudinal vehicle automation. All lateral maneuvers such as lane changes and lane keeping, as well as perception-based functions required for full autonomous driving, are outside the scope of the system. These functions are handled by the human driver or by separate supervisory systems during both VIL and on road experiments. This assumption reflects the design objective of the framework, which focuses on energy-efficient and safe longitudinal behavior rather than full autonomy.

In addition to the demonstrated energy-saving benefits, the integrated traffic-vehicle control framework also has meaningful environmental implications. Because energy consumption are strongly driven by acceleration events, idling during red lights, and frequent stop-and-go operations, the smoother speed trajectories produced by the RAED algorithm and the increased arrival-on-green rates enabled by the predictive bilinear signal controller can substantially reduce real-world energy consumption.

A key advantage of the proposed framework is its scalability for deployment across diverse urban corridors. The integrated traffic-vehicle control system can be transferred to new locations through a phased workflow in which simulation-based testing is first used to configure corridor-specific traffic demand and signal timing plans, followed by vehicle-in-the-loop evaluation to account for real vehicle dynamics and communication latency, and finally on-road field testing to validate performance in mixed traffic. Emerging digital twin and edge-computing technologies further streamline this process by enabling rapid reconstruction of corridor models and real-time coordination between infrastructure and CAVs, supporting practical and scalable adoption in different cities.

A limitation of the present study is that communication robustness was not evaluated through controlled degradation experiments. The RTMCS platform used here was previously characterized in Xu et al. [56], which reported that all 2,002 measured end-to-end V2I latencies were at most 0.9 s under real 4G urban field conditions, with a median measured latency of 0.5 s and a per-message algorithm execution time under 0.05 s. However, neither that study nor the present work measured packet-loss rate, communication range, stale-message frequency under degraded conditions, or the resulting impacts on RAED advisory availability and vehicle energy performance under deliberate V2I disruption. Future work should incorporate controlled latency injection and packet-loss testing in both VIL and field environments to quantify communication tolerance thresholds and to evaluate how degraded V2I information affects arrival-on-green rate, advisory speed feasibility, stop frequency, safety fallback frequency, and energy consumption.

A further limitation of this study is that the energy evaluation was conducted using a single vehicle platform: a 2017 Toyota Prius Prime plug-in hybrid operating in EV-only mode. The RAED coasting and energy model uses vehicle-specific road-load coefficients and vehicle mass. Therefore, the reported energy savings are specific

1031 to the tested vehicle configuration. The traffic signal control component is indepen-
1032 dent of vehicle type, but the vehicle-side eco-driving benefit depends on mass, rolling
1033 resistance, aerodynamic drag, powertrain efficiency, regenerative braking capability,
1034 and baseline driving behavior. Consequently, the reported energy-saving percentages
1035 should not be interpreted as directly generalizable to all vehicle classes. Future work
1036 should conduct a vehicle-class sensitivity analysis by re-parameterizing the RAED and
1037 energy models and by evaluating how vehicle mass, road load, grade, and powertrain
1038 characteristics affect the magnitude of eco-driving benefits.

1039 Based on the extensive testing results achieved through different experimental
1040 stages, we are proposing the following two possible future research approaches that
1041 would be able to achieve more robust and statistically reliable evaluation results for
1042 the deployment of this type of CAV technology:

- 1043 1. A large-scale, long-term daily operational on-road experiment needs to be con-
1044 ducted, which would allow the team to collect enough evaluation data that not
1045 only can cover all types of scenarios (i.e., different segment length, types of traf-
1046 fic signal controllers, roadway geometrics, AOG/AOR, and environmental traffic
1047 status) but can also provide a large sample size for each scenario.
- 1048 2. Combining the strengths of traffic simulations, VIL experiments, and on-road
1049 testing, we could first identify all the critical energy-sensitive factors through
1050 large-scale traffic simulations. Next, a relatively good sample size should be col-
1051 lected through well-designed (i.e., considering all the identified energy-sensitive
1052 factors) on-road testing to cover all types of real-world scenarios. Finally, after
1053 feeding all the on-road testing data into the VIL testing platform to replicate and
1054 expand the corresponding scenarios, and more “apples-to-apples” comparisons
1055 should be conducted in the future.

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1064 Data Availability

1065 Source data are provided with this paper.

1066 Competing Interests

1067 JY, WL, CW, TL, HX, AB, HL, YS, HW, AE, and JS have filed a nonprovisional
1068 patent application (US20250131819A1, pending) from Oak Ridge National Labora-
1069 tory, Oak Ridge, TN, 37830, USA. HX, JY, CW, TL, AB, WL, YS, HW, JS ave filed

1070 a nonprovisional patent application (US20250128708A1, pending) from Oak Ridge
1071 National Laboratory, Oak Ridge, TN, 37830, USA. The other authors declare no
1072 competing interests.

1073 Author Contributions

1074 The authors confirm their contributions to the paper as follows: study conception and
1075 design: JY, WL, CW, BC, TL, HW, and JS; data collection: JY, WL, CW, BC, TL,
1076 HW, HX, AB, PK, BG, YS, AE, QW, JC, and JS; analysis and interpretation of
1077 results: JY, WL, and CW; real-world implementation: JY, WL, CW, BC, TL, HW,
1078 HX, AB, PK, BG, YS, ARE, QW, and JS; draft manuscript preparation: JY, WL,
1079 CW, TL, HX, AB, PK, HW, BC, HL, BG, YS, AE, JC, QW, JS. All authors reviewed
1080 the results and approved the final version of the manuscript.

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